Study on Classroom Temperature Distribution in Hybrid Cooling System at Different Air Conditioner Setpoints

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***Abstract*—Air-conditioning (AC) is essential to maintain comfortable indoor environments, particularly in classrooms, resulting in higher energy consumption in educational buildings compared to neighboring countries. This study investigates the effectiveness of fan-assisted cooling as a hybrid strategy to reduce AC energy use. Experiments were conducted in two classrooms: Room 1 with a single fan and Room 2 with double fans, across AC setpoints from 26 deg C to 22 deg C. Hourly temperature and energy consumption data were collected and analyzed. Results show that fan-assisted cooling can significantly lower peak indoor temperatures and total energy use, with the magnitude of temperature reduction dependent on sensor location and airflow distribution. Notably, the double-fan configuration achieved the largest reductions, highlighting the importance of fan placement and room-specific airflow patterns. Daily energy analysis revealed that Room 1 achieved savings with AC and 1 fan at higher setpoints (26–24 deg C), while at lower setpoints (23–22 deg C), hybrid cooling sometimes consumed more energy than AC alone. In Room 2, the double-fan configuration resulted in consistently higher energy use across all setpoints, highlighting the influence of room-specific conditions and fan placement. These findings demonstrate that hybrid cooling strategies can enhance indoor comfort while reducing energy demand in educational buildings.**

Keywords—air-conditioning, hybrid cooling, indoor temperature, AC setpoin

I. Introduction

The rise in global temperatures and energy demand has leds to increasing energy consumption by air-conditioning (AC) units [1]. Factors such as building insulation, ventilation, AC usage, and the exchange dynamics between two environments heavily influence indoor temperatures and cooling loads [2]. High outdoor temperatures combined with inadequate heat transfer result in excessive energy usage [3]. Various strategies have been developed to improve energy efficiency including adjusting the AC setpoint [4]. Increasing the cooling setpoint from 22 °C to 27 °C can reduce the cooling load by 35.7% and electricity use by 44.6 MWh annually [5], with potential energy savings up to 60% while maintaining thermal comfort [6]. Another widely adopted approach is the combined use of AC and fans. Studies have shown that ceiling fans at low speed can maintain comfort while reducing energy consumption, particularly in tropical climates [7] with up to a 10% reduction when AC setpoints are raised to 25–27 °C in combination with fan operation [8]. To address the challenge of high energy consumption, while maintaining thermal comfort, this study evaluates the performance of AC-only and hybrid cooling (AC & fan) in classroom settings, analysing the effects of varying fan configurations and AC setpoints (T_{set}) on temperature distribution and energy consumption.

II. Methodology

A. Experimental Layout

The experiment was conducted in two classrooms, room 2A.10 and room 2A.11, located at the second floor at Phase 1 Building in Universiti Teknologi Brunei. Fig. 1 shows the floor plan of both classrooms. The first room is referred to as Room 1 (R1) indicating room with single fan whereas the latter is Room 2 (R2) indicating room with double fans. Fig. 2 shows the dashboard view of the monitoring system.

B. Experimental Parameters

In this work, the experiment parameters as in Table I is implemented during the experiment. The experimental parameters were defined to examine indoor temperature distribution under identical environmental conditions with fan configuration and air-conditioning T_{set} as the only varying parameters.

The experimental study was conducted for AC setpoints (T_{set}) ranging from 26 °C to 21 °C. For each fan speed level,

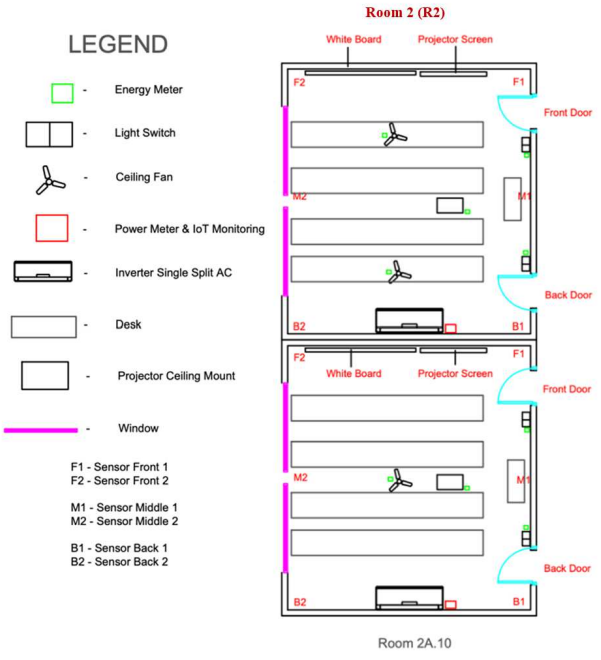


Fig. 1. Classrooms floor plan and sensor location.

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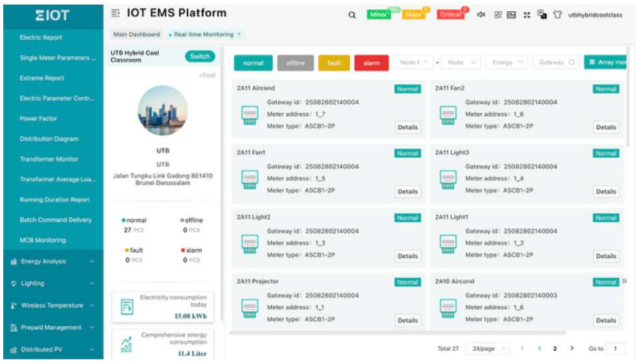


Fig. 2. Dashboard view of the monitoring system showing real-time data displaying the status of lighting, fans, and air conditioners for every classroom.

TABLE I. EXPERIMENT SETUP PARAMETERS

Parameters	Setting
Room Condition	Blind Closed and Lights ON
T_{set}	26°C to 21°C
Measurement duration	12 hours (6AM – 6PM)
Fan Configuration	Single and Double
Fan Speed Level	Level 5

in this case level 5, the complete test cycle required 12 days. Prior to AC operation, the fan was activated 15 minutes in advance to promote air circulation and stabilize indoor airflow conditions. Data collection for each T_{set} was carried out over two days using two identical rooms, with fan-assisted and non-assisted conditions alternated between rooms on consecutive days to minimize room-specific bias. Each experimental condition was continuously monitored for a duration of 12 hours. In this paper, the peak temperatures between fan-assisted and non-fan assisted operation is analyzed (ΔT_{max}) are analyzed using equation (1) where $T_{\max_S5}$ is the maximum temperature recorded at fan speed level 5 and $T_{\max_S0}$ is the maximum temperature recorded at fan speed level 0. A negative ΔT_{max} values indicate the fan operation reduced peak temperatures, representing a cooling benefit. A positive value indicates that fan operation increased peak temperatures.

$$\Delta T_{\max} = T_{\max_S5} - T_{\max_S0} \quad (1)$$

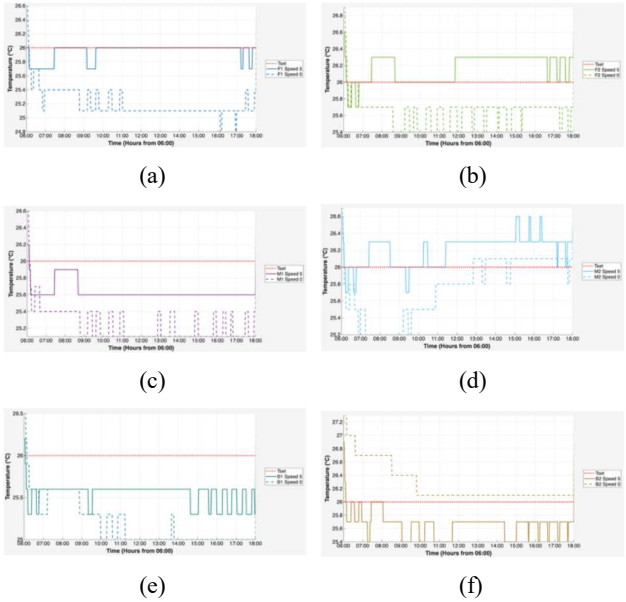
III. RESULTS AND DISCUSSION

The experimental findings obtained are presented and discussed in this section consisting of indoor temperature measurements under varying air-conditioning setpoints and fan configurations. To evaluate the influence of the hybrid cooling approach, the observed trends are analysed. This section is subdivided into two parts: single fan configuration and double fan configuration.

A. Single Fan Configuration

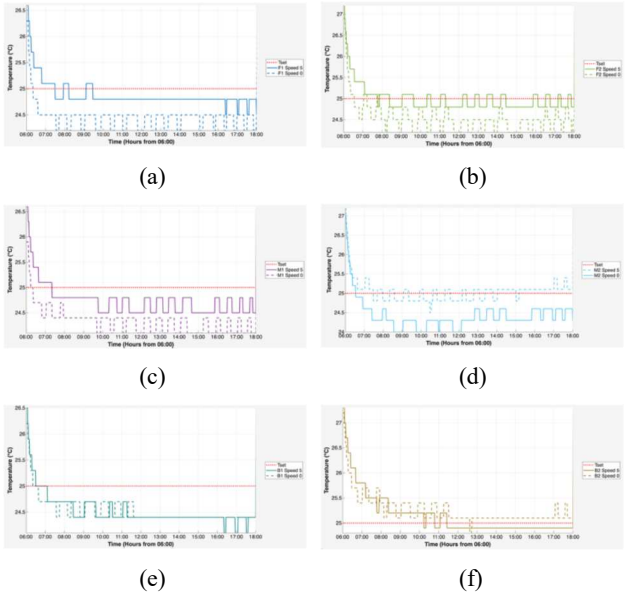
The results from six indoor sensors: F1, F2, M1, M2, B1 and B2 are presented and analysed for air-conditioning setpoints between 26°C and 21°C. Sensor readings at fan speed level 5 and 0 are compared under identical conditions to examine the impact of fan operation on indoor temperature. Fig. 3 to Fig.7 present the complete temperature profiles for single fan configuration, respectively.

Overall, the temperature distribution across locations varies with both AC setpoint and fan operation. For higher



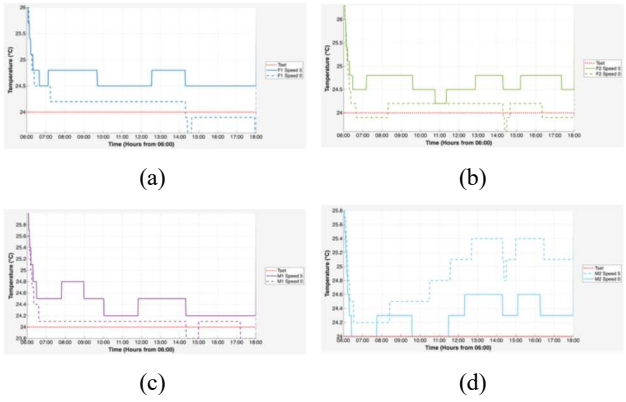
(a) (b) (c) (d) (e) (f)

Fig. 3. Comparison of indoor temperature measured for single fan configuration at fan speed 5 and 0 when T_{set} = 26°C by sensor (a) F1, (b) F2, (c) M1, (d) M2, (e) B1, (f) B2.



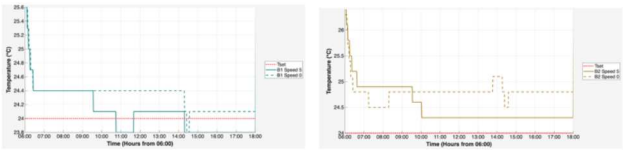
(a) (b) (c) (d) (e) (f)

Fig. 4. Comparison of indoor temperature measured for a single fan configuration at fan speed 5 and 0 when T_{set} = 25°C by sensor (a) F1, (b) F2, (c) M1, (d) M2, (e) B1, (f) B2.



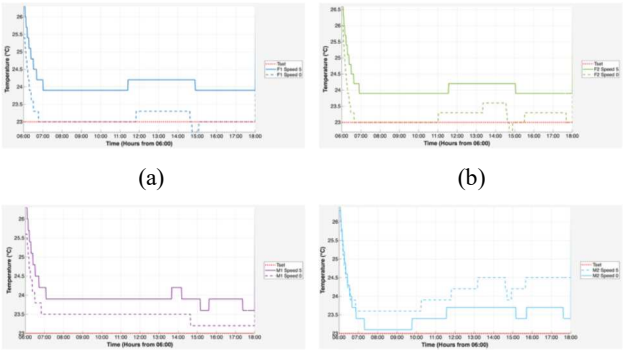
(a) (b) (c) (d)

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(e) (f)

Fig. 5. Comparison of indoor temperature measured for a single fan configuration at fan speed 5 and 0 when T_{set} = 24°C by sensor (a) F1, (b) F2, (c) M1, (d) M2, (e) B1, (f) B2.

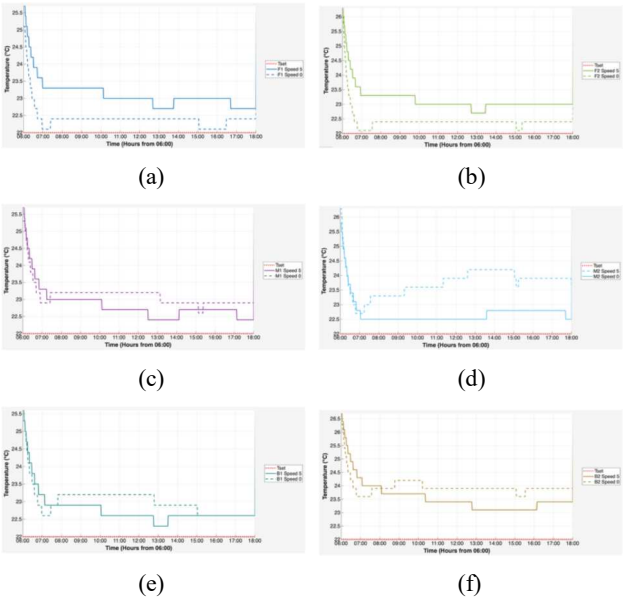


(a) (b)

(c) (d)

(e) (f)

Fig. 6. Comparison of indoor temperature measured for a single fan configuration at fan speed 5 and 0 when T_{set} = 23°C by sensor (a) F1, (b) F2, (c) M1, (d) M2, (e) B1, (f) B2.



(a) (b)

(c) (d)

(e) (f)

Fig. 7. Comparison of indoor temperature measured at single fan configuration at fan speed 5 and 0 when Tset = 22°C by sensor (a) F1, (b) F2, (c) M1, (d) M2, (e) B1, (f) B2.

setpoints (26–23 °C), the front and middle locations (F1, F2, and M1) generally record lower temperatures under fan speed 0 operation, while M2 consistently exhibits lower temperatures under fan speed level 5 operation at 26–22 °C. At the back locations, B1 and B2 initially experience higher temperatures before gradually approaching or falling below

TABLE II. MAXIMUM TEMPERATURE DIFFERENCES BETWEEN FAN-ASSISTED AND NON-ASSISTED CONDITIONS

Sensor Location	ΔT_{max}				
	26	*25*	*24*	*23*	*22*
F1	1.2	0.3	0	0.9	0.6
F2	0.6	0.3	0	0.6	0.3
M1	1.3	0.7	0.4	0.7	0.4
M2	-0.2	-0.2	0.1	0.1	-0.5
B1	0.9	0.3	0	0.6	0.3
B2	1	0.1	0.1	0.7	0.4

ΔT_{max} range	Color code
1.2 – 1.0	
0.9 – 0.7	
0.6 – 0.4	
0.3 – 0.1	
0	
-0.1 – -0.3	
-0.4 – -0.6	
-0.7 – -0.9	

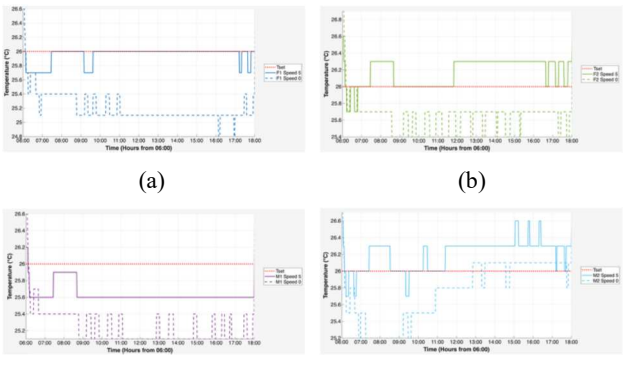
T_{set}, with only minor differences observed between speed 5 and speed 0 cases. At lower setpoints (24–22 °C), speed 5 operation becomes more effective at the middle and back locations (M1, M2, B1, and B2), although temperatures across all locations remain close to or slightly above T_{set} at 22 °C. These results indicate spatially dependent cooling behavior influenced by airflow distribution and fan operation.

Table II summarizes the differences between maximum temperatures at each measurement location for AC T_{set} for fan-assisted (speed 5) and non fan-assisted operation (speed 0) at varying AC setpoint for single fan configuration.

Based on the maximum temperature differences presented in Table II, the majority of ΔT_{max} values are positive (ranging from 0°C to 1.3°C) indicating higher maximum temperatures under the fan-assisted condition for most sensor locations and AC setpoints. The negative difference indicating temperature reduction occurred mostly at middle left (M2) of the classroom when T_{set} was at 26, 25 and 22°C. At 23 and 24°C, temperature raised but only at minimal value. The localized cooling benefit at M2 is potentially due to its placement within the fan's primary airflow path. However, the localized cooling was insufficient to offset the overall temperature increases observed elsewhere. One possible reason is that the fan motors contribute additional sensible heat to the conditioned space, even a small fan can generate 50-100W of a heat depending on its motor efficiency [9].

B. Double Fan Configuration

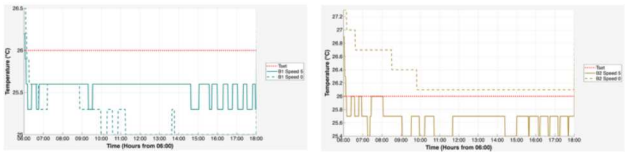
The same procedures are applied for double fan configuration where results from six indoor sensors: F1, F2, M1, M2, B1 and B2 are presented and analysed for air-conditioning setpoints between 26°C and 22°C at fan speed level 5 and level 0. Fig. 8 to Fig. 12 presents the raw temperature measurements from indoor sensor, F1 until B2 for fan speed 5 and 0 at AC setpoint of 26°C.



(a) (b)

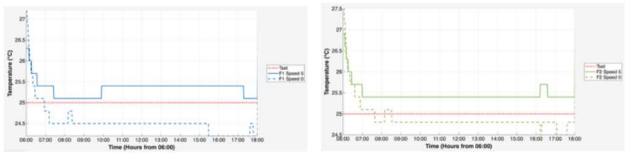
(c) (d)

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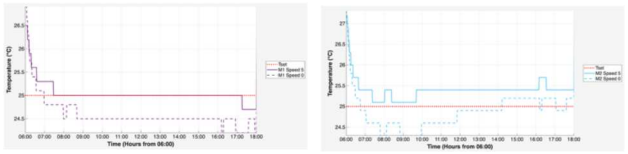


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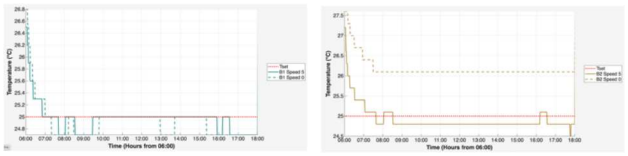
Fig. 8. Comparison of indoor temperature measured for double fan configuration at fan speed 5 and 0 when T_{set} = 26°C by sensor (a) F1, (b) F2, (c) M1, (d) M2, (e) B1, (f) B2.



(a) (b)

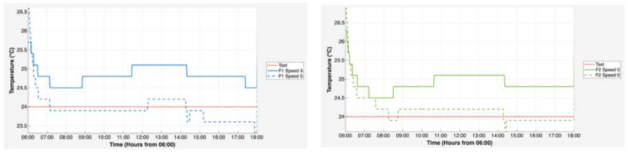


(c) (d)

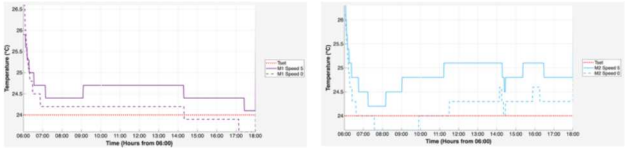


(e) (f)

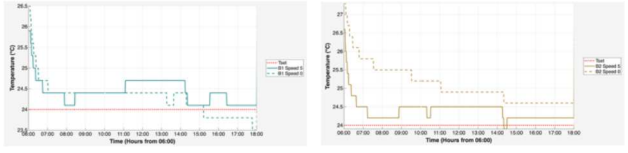
Fig. 9. Comparison of indoor temperature measured for double fan configuration at fan speed 5 and 0 when T_{set} = 25°C by sensor (a) F1, (b) F2, (c) M1, (d) M2, (e) B1, (f) B2.



(a) (b)



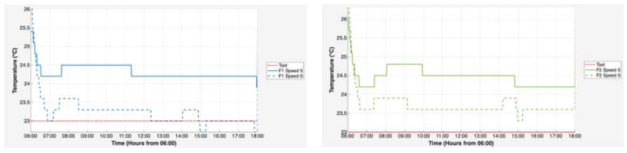
(c) (d)



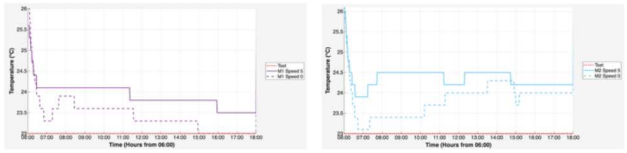
(e) (f)

Fig. 10. Comparison of indoor temperature measured for double fan configuration at fan speed 5 and 0 when T_{set} = 24°C by sensor (a) F1, (b) F2, (c) M1, (d) M2, (e) B1, (f) B2.

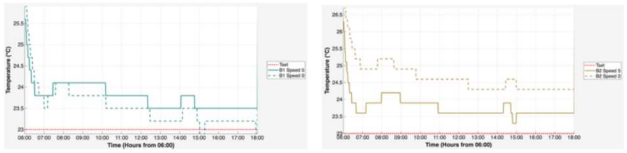
Across all setpoints (26–22 °C), front and middle locations (F1–M2) generally record lower temperatures under non–fan-assisted operation, particularly at higher setpoints. In contrast, fan-assisted operation becomes more effective at the back locations (B1 and B2), with B2 consistently exhibiting lower temperatures at speed 5, except at 26 °C where mixed



(a) (b)

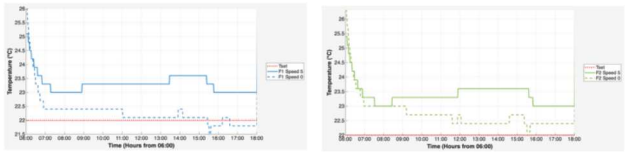


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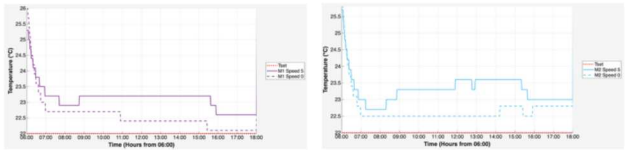


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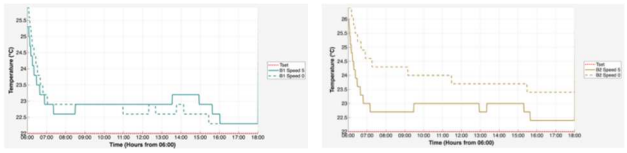
Fig, 11. Comparison of indoor temperature measured for double fan configuration at fan speed 5 and 0 when T_{set} = 23°C by sensor (a) F1, (b) F2, (c) M1, (d) M2, (e) B1, (f) B2



(a) (b)



(c) (d)



(e) (f)

Fig. 12. Comparison of indoor temperature measured for double fan configuration at fan speed 5 and 0 when T_{set} = 22°C by sensor (a) F1, (b) F2, (c) M1, (d) M2, (e) B1, (f) B2,

behavior is observed. These observations indicate that the effectiveness of fan-assisted cooling varies across different locations within the room, primarily influenced by the airflow patterns and distribution.

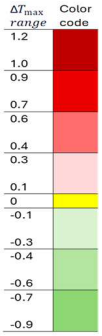
Table III summarizes the differences between maximum temperatures at each measurement location under fan-assisted (speed 5) and non fan-assisted operation (speed 0) at varying AC setpoint for double fan configuration.

Based on the results presented in Table III, almost all maximum temperature differences (ΔT_{max}) are negative, indicating that fan-assisted operation consistently reduces the peak indoor temperature across all sensor locations and AC setpoints. The largest temperature reduction is observed at sensor location F1 for all T_{set}. This shows that the fan was most effective at this location. In contrast, the smallest

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TABLE III. MAXIMUM TEMPERATURE DIFFERENCES BETWEEN FAN-ASSISTED AND NON-ASSISTED CONDITIONS

Sensor Location	ΔT_{max}				
	26	*25*	*24*	*23*	*22*
F1	-0.6	-0.9	-0.9	-0.6	-0.9
F2	-0.3	-0.6	-0.6	-0.3	-0.9
M1	-0.4	-0.4	-0.7	-0.4	-0.7
M2	-0.1	-0.1	0.2	-0.1	-0.1
B1	-0.3	-0.3	-0.6	-0.3	-0.6
B2	-0.4	-0.4	-0.7	-0.4	-0.4



reduction (ΔT_{max}= −0.1 °C) occurs mostly at sensor location M2 except at T_{set} = 24°C where there was a slight temperature increase indicating a relatively weaker influence of fan assistance at this location.

Comparing between single and double fan, double fan shows more temperature distribution improvement when compared without fan or with single fan. A more uniform temperature reduction is achieved where two fans provided greater total airflow which covers more areas of the room, reducing stagnant zones and interacting airflow patterns enhance air mixing and reduce thermal stratification. Hence, Table III shows cooling benefits across most sensor locations, compared to single-fan where cooling was limited to M2.

Table IV summarizes the total daily energy consumption (kWh) for AC-only and fan-assisted configurations (single and double fans) at AC setpoints between 26°C and 21°C for fan speed level 5.

Based on Table IV, the highest daily energy consumption under the AC-only configuration occurs at T_{set}= 22 °C for both Room 1 and Room 2. A similar trend is observed under hybrid cooling conditions, where the maximum energy consumption also occurs at 22 °C, for Room 1, whereas the highest energy occurs at T_{set}= 23 °C Room 2. For Room 1, at higher setpoints (26–24 °C), the AC-only configuration consistently consumes more energy than the AC & 1 fan configuration, indicating potential energy savings when a single fan is employed. However, at lower setpoints (23–22 °C), energy consumption increases substantially under AC + 1 fan, higher than AC only operation. In contrast, when two fans are used, the hybrid configuration exhibits relatively high energy consumption across almost all setpoints, suggesting diminishing energy-saving benefits at both higher and lower T_{set} values except at 22°C where energy was saved in hybrid mode.

Fig. 14 shows the variation in daily power consumption with T_{set}, where negative values denote energy savings and positive values indicate increased consumption. Room 1 exhibits the highest energy saving at the highest 26 °C (-1.48 kWh) whereas Room 2 shows reduced energy consumption of -0.31kW at the lowest setpoint, 22°C compared to AC-only and increased consumption for the remaining of higher T_{set}. The variation might be because of the AC operates. At higher setpoints for instance 26-24°C, the AC cycles intermittently to keep the temperature steady. The AC's cycling may be disrupted with the presence of fans in this situation and heat from motor may be added up, which could explain the rise in energy consumption. At low T_{set}, AC is likely to continuously running to keep the room cool, therefore the fans assisted in spreading the cool air around without interference with compressor cycling, helping AC to run more efficiently so energy consumption goes down.

TABLE IV. TOTAL DAILY ENERGY CONSUMPTION OF AC-ONLY AND FAN ASSISTED SYSTEMS AT VARIOUS AC SETPOINT

	AC only (kWh)		AC & 1 fan (kWh)			AC & 2 fans (kWh)			
T_{set}	R1	R2	Fan	AC	Total	F1	F2	AC	Total
26	**6.29**	5.24	0.28	4.53	**4.81**	0.37	0.31	6.19	6.87
25	**6.49**	6.65	0.29	5.66	**5.95**	0.36	0.31	6.26	6.93
24	**7.78**	6.76	0.29	7.13	**7.42**	0.36	0.32	7.2	7.88
23	8.72	8.07	0.28	9.43	9.71	0.36	0.31	8.58	9.25
22	9.05	**9.50**	0.29	11.28	11.57	0.36	0.32	8.51	**9.19**

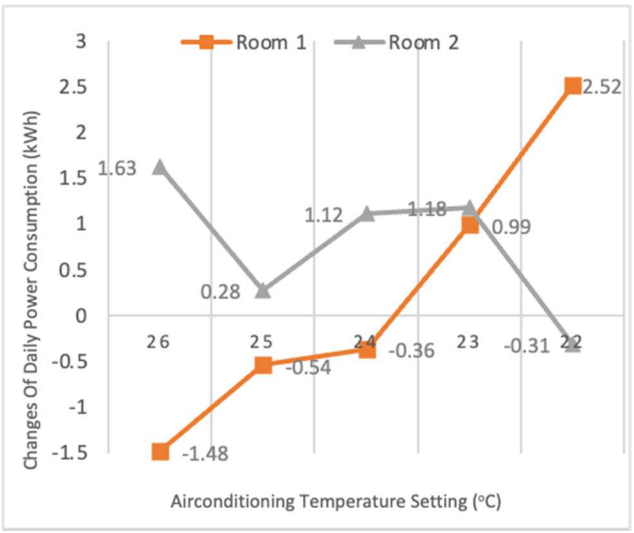


Fig. 13. Total of power consumption changes when using single fan, double fan in comparison with AC-only.

IV. CONCLUSION

Single-fan configuration showed localized cooling only at M2 (26, 25, 22°C), while double-fan achieved uniform temperature reduction across most sensor locations. These findings highlight that fan placement and airflow distribution critically determine cooling effectiveness. For energy consumption, Room 1 achieved savings with single fan at higher setpoints (26-24°C) but penalties at lower setpoints (23-22°C). In Room 2, double-fan resulted in consistently higher energy consumption across all setpoints, except at 22°C where marginal savings (0.31 kWh) were observed. This shows that increasing fan numbers does not guarantee energy savings and that room-specific configurations matter. These findings extend previous research on fan-assisted cooling [7],[8] by demonstrating that spatial temperature distribution and room-specific factors significantly influence energy outcomes. This study was conducted in unoccupied classrooms and focused on fan speed level 5; further research should examine occupancy effects and other fan speeds. Overall, fan-assisted hybrid cooling can be an effective strategy to enhance thermal comfort while optimizing energy use. Energy savings depend on setpoint and fan configuration, with single-fan providing benefits at higher setpoints. These findings provide practical guidance for designing energy-

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efficient cooling strategies in classrooms under tropical climates. As cooling demand rises, these findings support setpoint-based fan operation strategies that balance efficiency with comfort.

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